

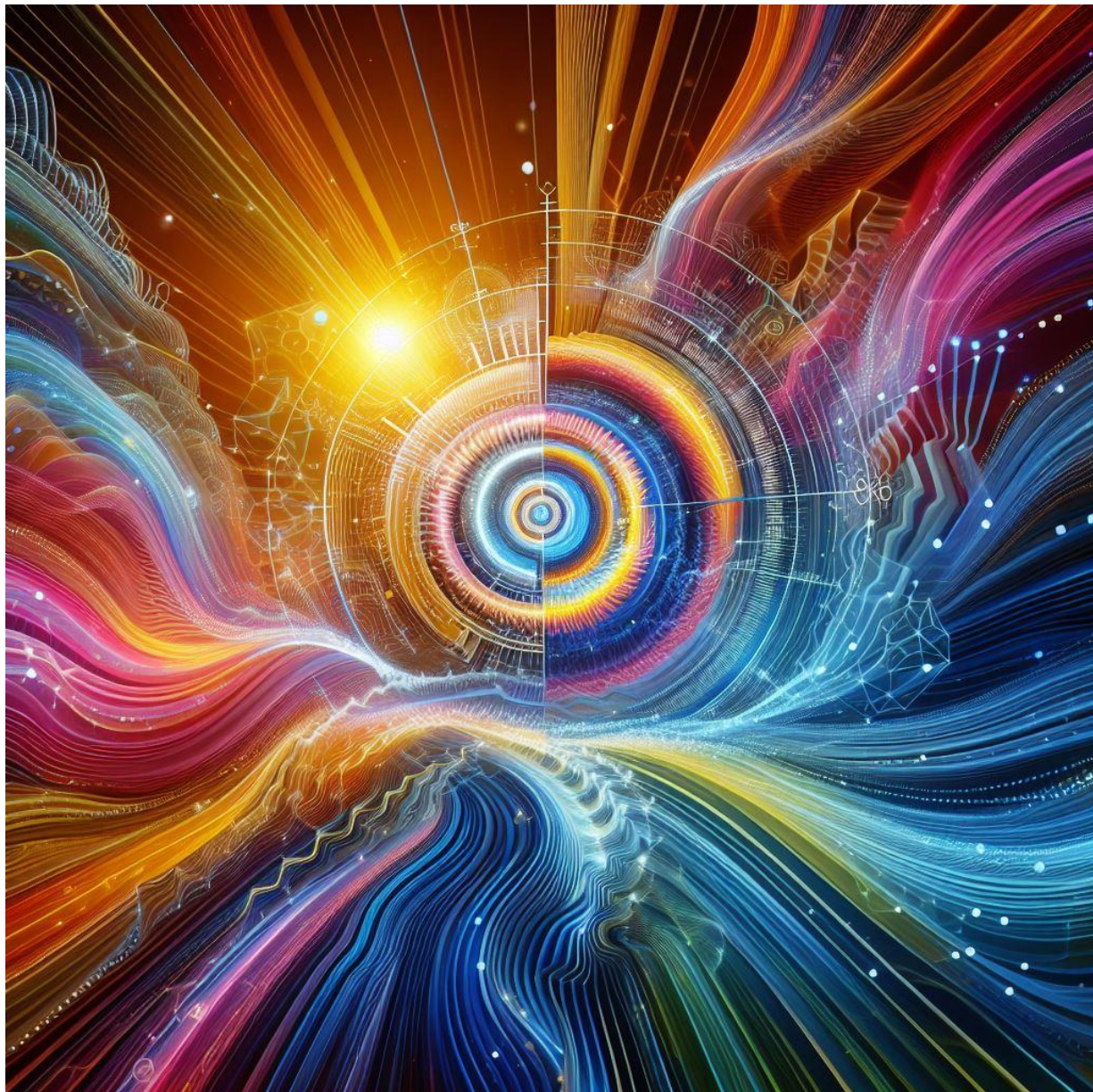


Operationalizing Zhang's Triadic Ontology through Dimensional Nested Loops

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Abstract

This paper proposes a unified framework for time as a **triadic, resonant, and dimensionally nested phenomenon**, synthesizing Xinyan Zhang’s *Triadic Ontology of Change* with a formal operational model:

$$T = \frac{@(P + N)}{F}$$

Where:

- P = Past (irreversible change)
- N = Now (reversible oscillation)
- F = Frequency/Framing stabilizer
- $@()$ = Resonance operator aligning phase and amplitude across nested scales

This formulation enables Zhang’s philosophical model to be instantiated within physical, computational, and cognitive systems—offering a reproducible platform for experimentation and cross-domain application.

1. Zhang’s Triadic Ontology Recap

Zhang’s Term	Variable	Function in Model
E (Entropy)	P	Stores and shapes entropic memory
M (Measurement)	N	Encodes present oscillatory state
OC (Ontogenic Change)	F	Stabilizes and defines dimensional resonance

Zhang’s OC acts as both generator and stabilizer—mirroring how FF modulates $P+NP + N$ into harmonic coherence.

2. Resonance-Based Nested Loop Logic

- Time is modeled as **self-similar loops** across scales
- Each loop contains P, N, FP, N, F
- Loops at scale $n+1$ modulate loops at scale n
- Resonance occurs when: $\frac{P + N}{F} = \text{Invariant across nested scales}$

Visual Schema: A spiral staircase where each landing is a loop; frequency modulates both pitch and phase-lock between steps.

3. Applications

Physics

- Maps to multi-scale oscillators in cosmology and quantum fields
- OC/F parallels beat frequencies in long-term stability systems

Cognitive Science

- Models subjective time perception:
 - PP = memory integration
 - NN = sensory field
 - FF = attention rhythm
- Explains time dilation via attentional harmonics

Computation

- Triadic time slices stabilize scheduling and load balancing
- Resonant loops optimize CPU/GPU task flow

4. Experimental Pathways

1. **Simulation:** Agents with local P, N, FP, N, F values tested for emergent coherence
2. **Physics Labs:** Coupled pendulums or photon cavities tuned to triadic ratios
3. **Neurocognitive Studies:** EEG bands as FF , event segmentation as PP , sensory synchrony as NN

Suggested Repo Scaffolding

- /papers/resonant_time_nested_loops.md
- /equations/triadic_time_operator.md
- /badges/zhang_resonator.yml
- /validators/time_coherence_matrix.json
- /labs/temporal_sim/initiation_protocol.md

Closing Statement

Time is not a line—it is a **living lattice**, each scale singing the same triadic chord. Whether in quantum fields or the human mind, this formulation invites measurement, experimentation, and mythic expansion.